

# Assignment\_2 Report

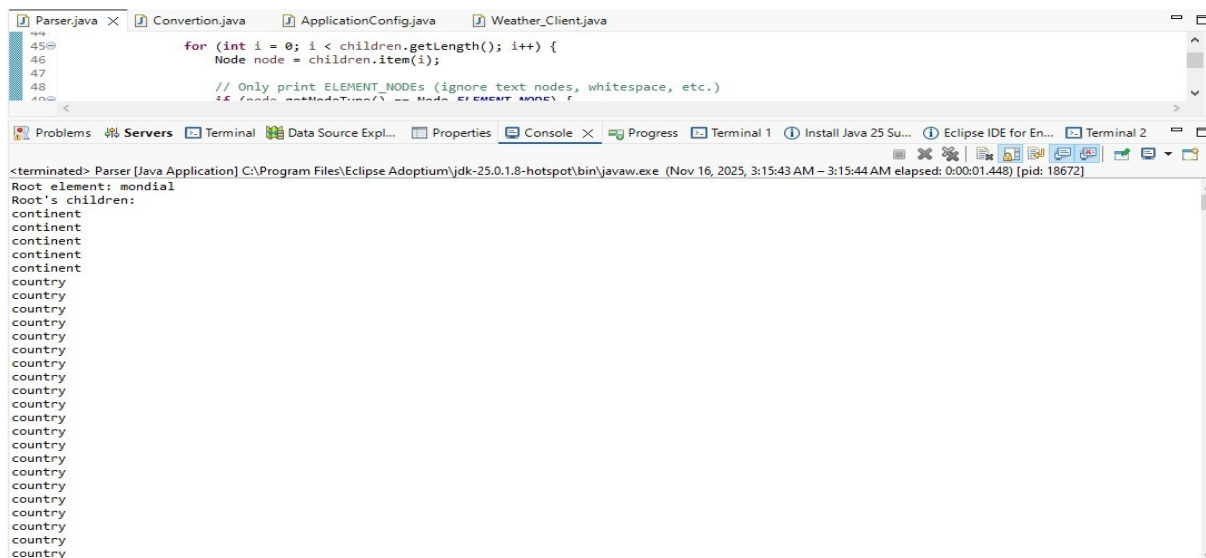
## Νικολέτα Ξενάκη

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\*I used JavaSE21 for spring boot applications and my installed: jdk-25.0.1.8-hotspot for the other applications.

## Question1

I)

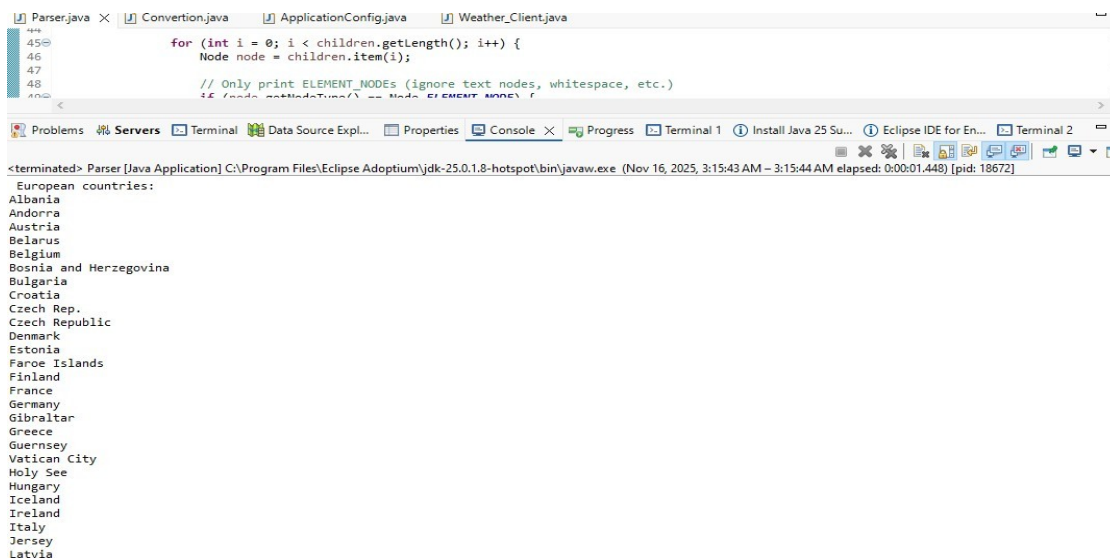


The screenshot shows the Eclipse IDE with several Java files open: Parser.java, Conversion.java, ApplicationConfig.java, and Weather\_Client.java. The Console view at the bottom displays the output of a Java application. The output starts with a terminated message and then lists the children of a root element named 'mondial'. The children are categorized into 'continent' and 'country'. The 'continent' entries are repeated five times, and the 'country' entries are repeated ten times.

```
<terminated> Parser [Java Application] C:\Program Files\Eclipse Adoptium\jdk-25.0.1.8-hotspot\bin\javaw.exe (Nov 16, 2025, 3:15:43 AM – 3:15:44 AM elapsed: 0:00:01.448) [pid: 18672]
Root element: mondial
Root's children:
continent
continent
continent
continent
continent
country
country
country
country
country
country
country
country
country
country
country
country
country
country
country
country
```

and many more...

ii) Print (using XPath) the names of all European countries:

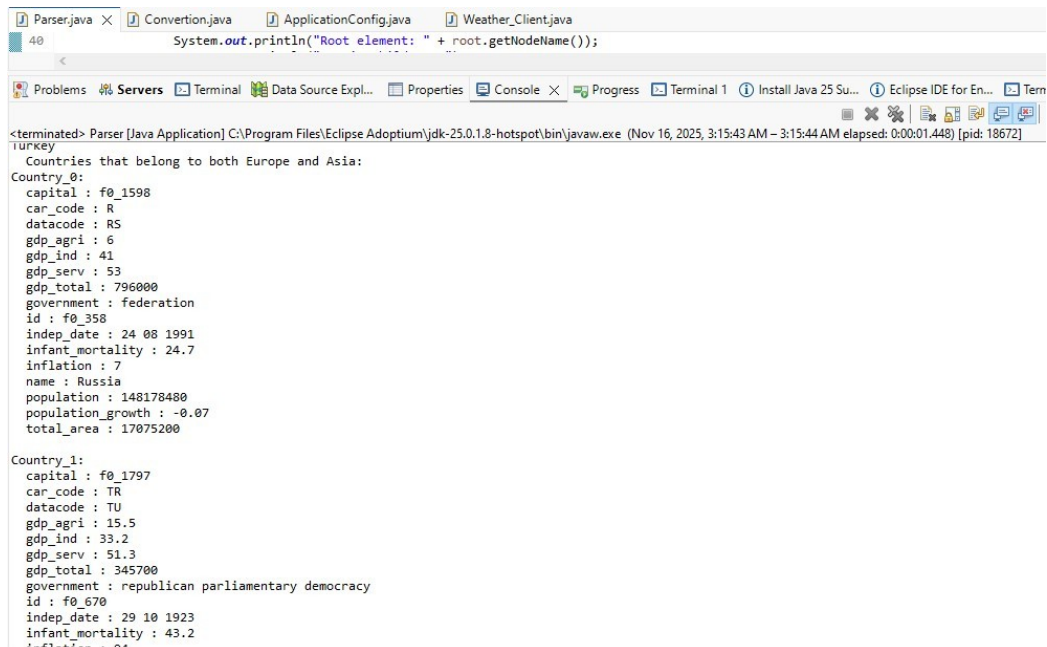


The screenshot shows the Eclipse IDE with the same Java files as before. The Console view at the bottom displays the output of a Java application. The output starts with a terminated message and then lists the names of all European countries. The list includes: Albania, Andorra, Austria, Belarus, Belgium, Bosnia and Herzegovina, Bulgaria, Croatia, Czech Rep., Czech Republic, Denmark, Estonia, Faroe Islands, Finland, France, Germany, Gibraltar, Greece, Guernsey, Vatican City, Holy See, Hungary, Iceland, Ireland, Italy, Jersey, and Latvia.

```
<terminated> Parser [Java Application] C:\Program Files\Eclipse Adoptium\jdk-25.0.1.8-hotspot\bin\javaw.exe (Nov 16, 2025, 3:15:43 AM – 3:15:44 AM elapsed: 0:00:01.448) [pid: 18672]
European countries:
Albania
Andorra
Austria
Belarus
Belgium
Bosnia and Herzegovina
Bulgaria
Croatia
Czech Rep.
Czech Republic
Denmark
Estonia
Faroe Islands
Finland
France
Germany
Gibraltar
Greece
Guernsey
Vatican City
Holy See
Hungary
Iceland
Ireland
Italy
Jersey
Latvia
```

and some more..

iii)



```
Parser.java  Conversion.java  ApplicationConfig.java  Weather_Client.java
40  System.out.println("Root element: " + root.getNodeName());

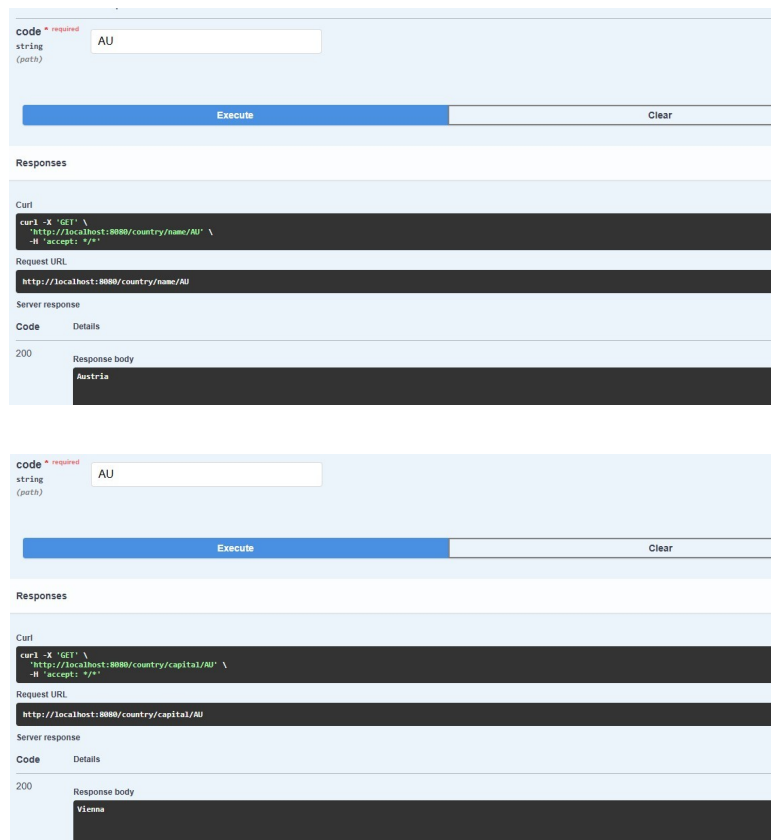
<terminated> Parser [Java Application] C:\Program Files\Eclipse Adoptium\jdk-25.0.1.8-hotspot\bin\javaw.exe (Nov 16, 2025, 3:15:43 AM - 3:15:44 AM elapsed: 0:00:01.448) [pid: 18672]
Turkey
Countries that belong to both Europe and Asia:
Country_0:
capital : f0_1598
car_code : R
datacode : RS
gdp_agri : 6
gdp_ind : 41
gdp_serv : 53
gdp_total : 796000
government : federation
id : f0_358
indep_date : 24 08 1991
infant_mortality : 24.7
inflation : 7
name : Russia
population : 148178480
population_growth : -0.07
total_area : 17075200

Country_1:
capital : f0_1797
car_code : TR
datacode : TU
gdp_agri : 15.5
gdp_ind : 33.2
gdp_serv : 51.3
gdp_total : 345700
government : republican parliamentary democracy
id : f0_670
indep_date : 29 10 1923
infant_mortality : 43.2
inflation : 0.0
```

iv) The european\_countries.json is created in the root dir after running the Parser.java

## Question2

I) Παράδειγμα με την Austria (AU) με χρήση του διψήφιου κωδικού της και με το όνομα δύο ποταμών:



code \* required  
string (path)  
AU

Execute Clear

Responses

curl -X 'GET' \n "http://localhost:8080/country/name/AU" \n -H 'accept: \*/\*' \n

Request URL  
http://localhost:8080/country/name/AU

Server response

| Code | Details                  |
|------|--------------------------|
| 200  | Response body<br>Austria |

code \* required  
string (path)  
AU

Execute Clear

Responses

curl -X 'GET' \n "http://localhost:8080/country/capital/AU" \n -H 'accept: \*/\*' \n

Request URL  
http://localhost:8080/country/capital/AU

Server response

| Code | Details                 |
|------|-------------------------|
| 200  | Response body<br>Vienna |

code \* required

string

(path)

AU

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8080/country/political_system/AU' \
  -H 'accept: */*'

```

Request URL

http://localhost:8080/country/political\_system/AU

Server response

| Code | Details                                              |
|------|------------------------------------------------------|
| 200  | <div>Response body</div> <div>federal republic</div> |

code \* required

string

(path)

AU

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8080/country/population/AU' \
  -H 'accept: */*'

```

Request URL

http://localhost:8080/country/population/AU

Server response

| Code | Details                                     |
|------|---------------------------------------------|
| 200  | <div>Response body</div> <div>8023244</div> |

code \* required

string

(path)

AU

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8080/country/total_area/AU' \
  -H 'accept: */*'

```

Request URL

http://localhost:8080/country/total\_area/AU

Server response

| Code | Details                                   |
|------|-------------------------------------------|
| 200  | <div>Response body</div> <div>93859</div> |

name \* required

string

(path)

Blue Nile

Execute

Clear

Responses

Curl

```
curl -X 'GET' \
  'http://localhost:8080/river/outfall/BlueNile' \
  -H 'accept: */*'

```

Request URL

http://localhost:8080/river/outfall/BlueNile

Server response

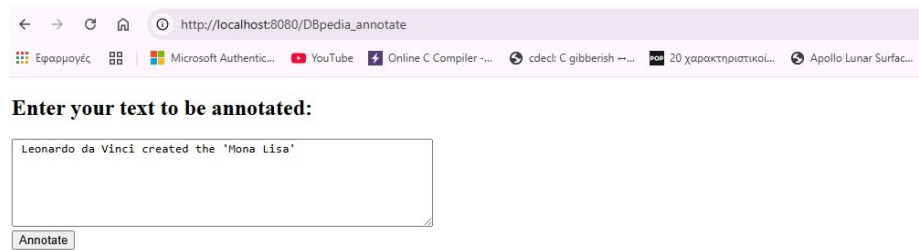
| Code | Details                                  |
|------|------------------------------------------|
| 200  | <div>Response body</div> <div>Nile</div> |



ii) Dbpedia annotation:

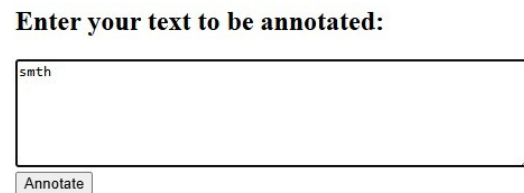
To connect the frontend of index.html with the backend I used Thymeleaf and in the pom.xml I added:

```
<dependency>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-thymeleaf</artifactId>
</dependency>
```



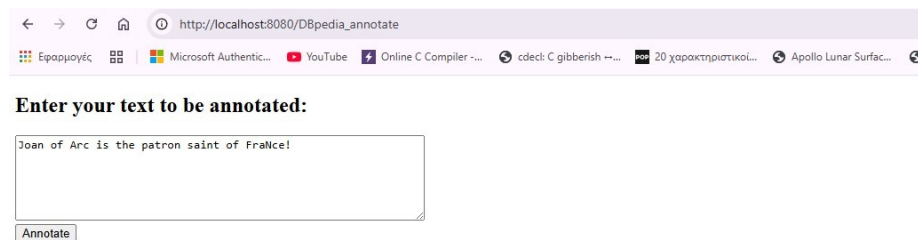
A screenshot of a web browser at the URL `http://localhost:8080/DBpedia_annotate`. The page has a title bar with several icons. Below the title bar, there is a heading "Enter your text to be annotated:". Underneath this heading is a text input field containing the text "Leonardo da Vinci created the 'Mona Lisa'". Below the input field is a button labeled "Annotate".

**Annotated Result:**  
[Leonardo da Vinci](#) created the '[Mona Lisa](#)'



A screenshot of the same web application. The heading "Enter your text to be annotated:" is present. The text input field now contains the text "smth". Below the input field is a button labeled "Annotate".

**Annotated Result:**  
`smth`



A screenshot of the same web application. The heading "Enter your text to be annotated:" is present. The text input field now contains the text "Joan of Arc is the patron saint of France!". Below the input field is a button labeled "Annotate".

**Annotated Result:**  
Joan of Arc is the [patron saint](#) of [France](#)!



ii) Client who makes weather requests for a city:

```
Problems Servers Terminal Data Source Expl... Properties Console X Progress Terminal 1 Install Java 25 Su... Eclipse IDE for En... Terminal 2
<terminated> Weather_Client [Java Application] C:\Program Files\Eclipse Adoptium\jdk-25.0.1.8-hotspot\bin\javaw.exe (Nov 16, 2025, 3:06:38 AM - 3:06:43 AM elapsed: 0:00:05.219) [pid: 15624]
Enter city: Heraklion
Weather XML: <?xml version="1.0" encoding="UTF-8"?>
<current>
<city id="8133920" name="Heraklion">
<coord lon="25.0929" lat="35.2445">
</coord>
<country>GR</country>
<timezone>7200</timezone>
<sun rise="2025-11-16T04:55:39" set="2025-11-16T15:13:24">
</sun>
</city>
<temperature value="12.7" min="11.68" max="13.01" unit="celsius">
</temperature>
<feels_like value="12.04" unit="celsius">
</feels_like>
<humidity value="77" unit="%">
```

```
<terminated> Weather_Client [Java Application] C:\Program Files\Eclipse Adoptium\jdk-25.0.1.8-hotspot\bin\javaw.exe (Nov 16, 2025, 3:05:46 AM - 3:05:53 AM elapsed: 0:00:06.778) [pid: 16640]
Enter city: Paris
Weather XML: <?xml version="1.0" encoding="UTF-8"?>
<current>
<city id="2988507" name="Paris">
<coord lon="2.3488" lat="48.8534">
</coord>
<country>FR</country>
<timezone>3600</timezone>
<sun rise="2025-11-16T07:01:20" set="2025-11-16T16:09:41">
</sun>
</city>
<temperature value="10.79" min="9.1" max="11.43" unit="celsius">
</temperature>
<feels_like value="10.35" unit="celsius">
</feels_like>
<humidity value="93" unit="%">
```

**THE END**  
**Thank you for your time!**